

Xiyuan GUO

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EDUCATION

University of Virginia

Charlottesville, VA — Sept 2024 – May 2028 (Expected)

Degree: Double Major in Behavioral Neuroscience and Computer Science — GPA: 3.89/4.00

PROFESSIONAL EXPERIENCE

Research Software Engineer, UVA (Prof. Liqin Ke's Group)

Charlottesville, VA — Oct 2025 – Present

- Applied quantum mechanics principles to engineer complex Density Functional Theory (DFT) reconstruction algorithms, bridging theoretical atomic modeling with robust software implementations to accurately catalog electronic material properties.
- Engineered a cloud-native machine learning pipeline utilizing Go and Next.js, enabling global academic collaborators to securely share specialized datasets and execute pre-trained predictive models.
- Created immersive, browser-based data visualizations using Three.js and JSmol, generating interactive 3D atomic models and Kiviat diagrams to facilitate the real-time structural inspection and validation.

Machine Learning Assistant, Yale (Prof. Mengxia Liu's Group)

New Haven, CT — Feb 2026 – Present

- Formulated a deep unrolled ADMM framework to solve ill-posed inverse problems in computational imaging, reconstructing multi-dimensional data encompassing spatial, spectral, and full-Stokes polarization domains.
- Engineered a specialized PyTorch pipeline utilizing a custom convolutional U-Net architecture as a learned proximal operator, specifically designed to mitigate non-Gaussian hardware sensor noise and yielding a 4.5 dB increase in PSNR.
- Implemented rigorous physics-informed constraints within deep learning optimization loops, mathematically grounding vectorial Stokes parameter representations utilizing complex numbers, trigonometric formulations, and 3D Poincaré sphere mappings.

Quantitative Researcher, UVA (Prof. Xin Cynthia Tong's BAL Lab)

Charlottesville, VA — Apr 2026 – Present

- Engineered a mathematical framework for multivariate longitudinal missing data imputation that enforces the Law of Total Variation, mitigating structural covariance bias inherent in unconstrained predictive machine learning models.
- Formulated a Lagrange-constrained optimization engine, deriving custom matrix calculus gradients from first principles to mathematically project non-parametric Random Forest estimates onto a positive-semidefinite (PSD) manifold.
- Architected a custom high-performance C++17 gradient descent solver via RcppArmadillo to iteratively evaluate dual Lagrangian multipliers, ensuring the final global variance of the imputed dataset converges to strict theoretical asymptotic expectations.

Network Science Intern, UVA Biocomplexity Institute

Charlottesville, VA — May 2026 – Present

- Formulated massive-scale network simulations to dynamically model the stochastic spread of viral pathogens (e.g., COVID-19 and Ebola), implementing SEIR compartmental cascade models to capture heterogeneous transmission dynamics.
- Engineered attention-based Graph Neural Network (GNN) architectures to predict multi-hop epidemic propagation pathways, leveraging spatiotemporal message-passing frameworks to extract predictive topological features from dynamic contact networks.

PROJECTS

Creator & Lead Maintainer, difi (Go, Bubble Tea)

Dec 2025 – Present

- Engineered a high-performance Go-based Git diff terminal user interface utilizing the Bubble Tea framework to streamline version control workflows, achieving official Homebrew Core inclusion, 342+ GitHub stars, and a #5 ranking on the Hacker News front page.

Creator & Lead Developer, sanjaya (Python, Playwright, Scrapy)

May 2026 – Present

- Engineered an automated academic literature extraction pipeline utilizing the OpenAlex Academic Graph API, Scrapy, and Playwright to extract bilingual data, exporting structured .CSV and .JSON datasets for downstream interdisciplinary research applications.

AWARDS & HONORS

- **Double Hoo Research Award:** Secured competitive grant funding to drive independent, interdisciplinary computational research.
- **Academic Honors:** USA Computing Olympiad (USACO) Silver Division; UVA Dean's List of Distinguished Students.

CERTIFICATIONS & SKILLS

- **Math & Science:** Quantum Mechanics, Density Functional Theory (DFT), Linear & Tensor Algebra, Real Analysis, ODEs
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, JAX, Scikit-Learn, XGBoost, Pandas, NumPy, OpenCV, RcppArmadillo
- **DevOps & Cloud:** Linux (RHEL/Fedora), AWS (EC2, S3), Docker, Kubernetes, CUDA, Slurm HPC, Git, CI/CD, Supabase, Sway, Nvim
- **Frameworks & Web:** React, Next.js, SvelteKit, Node.js, FastAPI, Bubble Tea, GraphQL, REST APIs, TailwindCSS
- **Languages:** Python, C/C++, Rust, Go, TypeScript, JavaScript, R, Julia, SQL, Lua, Bash, Java, L^AT_EX
- **Certifications:** AWS Certified AI Practitioner, AWS Certified Cloud Practitioner